GRADE 1 · BLOCK CODING

Start and Go

Run the blocks to find Bit's number

Name: .... Date: ....

★ **I can read code and find the number Bit lands on.**

Read the code

Bit moves on a number path. The green start block tells Bit which number to begin on. Each blue forward block moves Bit that many steps along the path. Run the blocks from the TOP, one at a time, to find the number Bit lands on.

Bit's number path

0	1	2	3	4	5

Code A

start at 0

forward 2

forward 1

1 Run Code A

Start on 0. Run each block in order: forward 2, then forward 1. What number does Bit land on? Write the number.

Code B

start at 1

forward 3

2 Predict, then run

PREDICT first: what number will Bit land on? Write your guess. Then run Code B from 1 and check. Were you right?

3 You try — change a block

In Code B, change forward 3 to forward 2. Predict the new landing number, then run it to check. What number now?

Big idea

Code is a list of blocks that run in order from the top. The start block goes first to set Bit's number, then each move block runs one at a time.

TEACHER GUIDE · GRADE 1 · BLOCK CODING

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Quick facilitation notes & answer key

What this worksheet teaches

Bit the robot stands on a number line from 0 to 5. The student reads a short list of instructions (called "blocks") from top to bottom, one at a time, to work out which number Bit ends up on — then changes one instruction and finds the new answer. No coding experience needed: if you can follow a recipe step by step, you can teach this page.

Before you start (no computer needed)

A paper-and-pencil activity — no app, tablet, or computer to set up. Just print the worksheet and hand out pencils. Optional: tape a big number line (0 to 6) on the floor so a child can walk out each instruction.

How to lead it (about 20 minutes)

1. Show them first (you do). Point to "Code A" and read it aloud, acting it out: "Start at 0 — stand on 0. Forward 2 — hop to 2. Forward 1 — hop to 3. Bit lands on 3." One step per instruction, top to bottom.
2. Do one together (we do). As a class, do Exercise 1 (lands on 3). For Exercise 2, have students GUESS Code B's answer first, then follow the steps to check (lands on 4).
3. Let them try alone (you do). In Exercise 3 they change "forward 3" to "forward 2", guess, then check (lands on 3).
4. Wrap up: "We read instructions from the top, one at a time. The first one tells Bit where to stand; the rest tell Bit how to move."

Answer key (these are the verified correct answers)

Exercise 1 — Code A: start on 0, forward 2 → 2, forward 1 → 3. Bit lands on 3.

Exercise 2 — Code B: start on 1, forward 3 → 4. Bit lands on 4.

Exercise 3 — after changing to forward 2: start on 1 → 3. Bit lands on 3.

Why order matters: the first instruction sets where Bit starts; a "forward" before it would have no number to move from. After that, instructions run one at a time, top to bottom.

If students get stuck — and ways to adjust

Watch for: counting the starting number as a move. "Start at 0" just means "stand on 0" — not a hop. The first hop is the first "forward".

Make it easier: have the child walk each instruction on a floor number line, pausing on each number.

Make it harder: change the start to "start at 2" and ask what Code A lands on now (5).

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

In plain terms: students follow a set of step-by-step instructions, in order, to get a result.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions someone else wrote, change one, and explain what the change did.

You'll know it worked when

The student follows the instructions in order and writes the correct landing number for each code.